A low-cost calibration workflow for the LC-Mare recorder: electro-mechanical model and reciprocity in a fixed tube and a fixed frame

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Abstract. Low-cost acoustic recorders produce science-grade data only if the sensitivity of the whole recording chain is known, but the established calibration methods are costly or limited in frequency. Here a calibration workflow for the low-cost recorder LC-Mare with SQ17-00 hydrophones is presented that combines an electro-mechanical model of the hydrophone with two reciprocity measurements, both with fixed arrangements of three transducers that are never moved during a measurement. The model, a two-mode equivalent circuit, takes most of its parameters from the electrical impedance. A water-filled PVC pipe of 1 m length, 63 mm outer diameter and 4.7 mm wall provides the primary calibration from about 36 Hz to 736 Hz, with a residual error of 0.14 dB at 170 Hz. A frame of flooded PVC tubes of 20 mm diameter and 1 m length, lowered to 10 m with the three recorders hung 15 cm below it and 56.3 cm apart, provides the free-field reciprocity from 2.2 kHz to 174 kHz with gated FM sweeps. The echoes of the frame tubes, which cannot be gated out, bias the sensitivity by at most 0.15 dB. A joint fit of the model to impedance, tube and frame bridges the gap between the two measurements and, in a Monte Carlo simulation, gives the sensitivity within 0.17 dB from 10 Hz to 170 kHz and the phase within 1.8°, whereas the model fitted to impedance and tube alone leaves errors of up to 3.26 dB and 16.3° near the resonances of the hydrophone. One could therefore recommend the tube for the low frequencies and the frame for the high frequencies, with the model serving to join them rather than to replace either.

I. INTRODUCTION

Passive acoustic monitoring relies increasingly on low-cost recorders, which are built from commercial hydrophones, audio converters and microcontrollers, and which are deployed in numbers that commercial instruments would not allow. The Low-Cost Marine Acoustic Recording system LC-Mare is such a recorder, with an SQ17-00 hydrophone of Sensortech (Canada), a differential preamplifier and the converter TLV320ADC6140 of Texas Instruments (Zimmer, 2026, Sec. 10.5). The data of these recorders are comparable with those of other instruments only if the sensitivity of the whole recording chain, from the sound pressure at the hydrophone to the digital sample, is known. Data sheets give nominal sensitivities that may deviate by several decibels, and the gains of preamplifiers and converters carry their own tolerances. A calibration of the complete system is therefore a pre-requisite, and it should give the phase as well as the amplitude, as direction finding and intensity estimates combine channels. For a low-cost system, the calibration itself should be affordable and simple enough to be repeated before and after each campaign.

The established methods address different frequency ranges at different costs. The free-field reciprocity and the comparison calibration in a tank (IEC 60565-1, 2020) require pulses that end before the first reflection arrives, which limits the lowest usable frequency to a few kilohertz, whereas many sounds of interest, from fin whale calls at 20 Hz to the sounds of fishes and of shipping, lie below. Low-frequency methods enclose the hydrophone in a small coupler, in which the sound pressure is uniform (Bobber, 1970; IEC 60565-2, 2019). A pistonphone requires a coupler that fits the housing, works in air, and must itself be traced to a primary standard, whereas the coupler reciprocity calibration in a water-filled chamber avoids a calibrated source, but in its usual form exchanges the transducers between measurements. Neither family covers the band from tens of hertz to beyond the first resonance of a hydrophone near 80 kHz.

Here a workflow is presented that covers this band with a minimum of equipment by combining modelling and measurement. The response of the hydrophone is described by an electro-mechanical model whose parameters are estimated from three kinds of measurements: the electrical impedance, a reciprocity calibration in a closed water-filled pipe for the low frequencies, and a reciprocity calibration with a fixed frame in the sea for the high frequencies. In both reciprocity measurements, the three transducers remain in a fixed arrangement, so that no distance changes between the measurements that are combined. The model joins the two bands, bridges the gap between them, and constrains the phase where the geometry alone cannot. The theory is summarized in Sec. II, the equipment and the workflow in Sec. III, and the expected performance in Sec. IV.

II. THEORY

A. Electro-mechanical model of the hydrophone

The SQ17-00 is described by a two-mode equivalent circuit (Zimmer, 2026, Appendix A.3), in which the clamped capacitance $C_H(1-c_1-c_2)$ is shunted by two motional branches, one for each mode. With the admittance of the motional branches

$$Y_m = \sum_{k=1}^{2} \left(R_k + \mathrm{i}\omega L_k + \frac{1}{\mathrm{i}\omega C_k} \right)^{-1}, \qquad C_k = c_k C_H, \quad L_k = \frac{1}{\omega_k^2 C_k}, \quad R_k = \frac{1}{\omega_k C_k Q_k} \tag{1}$$

the electrical impedance and the open-circuit sensitivity are

$$Z = \frac{1}{Y_m + \mathrm{i}\omega C_H (1-c_1-c_2)}, \qquad M = \frac{M_0}{\sqrt{1+(d_c f/f_1)^2}} \, \frac{Y_m Z}{c_1+c_2} \tag{2}$$

where C_H is the low-frequency capacitance, c_k the coupling factors, $f_k = \omega_k/(2\pi)$ the resonance frequencies and Q_k the quality factors of the two modes, M_0 the low-frequency sensitivity, and d_c a diffraction constant that describes the loss of sensitivity as the hydrophone ceases to be small. At low frequencies, $Y_m Z \to c_1 + c_2$ and $M \to M_0$. The same circuit describes impedance and sensitivity, so that C_H, f_k, Q_k and c_k follow from an electrical measurement, whereas M_0 and d_c have no electrical signature and require an acoustic measurement. This division is the basis of the workflow: the impedance gives the shape of the response, the tube gives its level, and the frame checks and refines the shape where the acoustics departs from the circuit.

B. Sound field in a closed pipe

Below the cutoff of the first cross mode, $1.841\, c_w/(2\pi a)$ or about 16 kHz for rigid walls of the inner radius a = 26.8 mm of the pipe, only plane waves propagate, and the sound pressure is uniform across the pipe. For a small source of volume velocity Q at the distance x_s from one end cap of a pipe of length L with rigid caps, the sound pressure at the distance x_r from the same cap is (Zimmer, 2026, Appendix A.3)

$$p(x_r) = -\mathrm{i}\, \frac{\rho c\, Q}{A \sin kL} \cos(k x_<) \cos(k(L - x_>)) \tag{3}$$

where $x_<$ and $x_>$ are the smaller and the larger of x_s and x_r, A is the inner cross-section, ρ the density and c_w the sound speed of free water, c the speed of the tube wave, $k = 2\pi f/c$ the wavenumber, and the time dependence is $e^{\mathrm{i}\omega t}$. Eq. (3) is symmetric in source and receiver, which expresses the reciprocity of the pipe. In the frequency range of the pipe, a hydrophone of 25.4 mm length is small compared with the wavelength and acts as a volume source irrespective of its orientation.

An elastic wall widens under the sound pressure and adds its compliance to that of the fluid, which lowers the speed of the tube wave to (Korteweg, 1878)

$$c = \frac{c_f}{\sqrt{1+\varepsilon}}, \qquad \varepsilon = \frac{2K}{Y}\left(\frac{b^2+a^2}{b^2-a^2}+\nu\right) \tag{4}$$

where c_f is the sound speed and K the bulk modulus of the free fluid, b the outer radius of the pipe, and Y and ν are Young's modulus and Poisson's ratio of the wall. For water, K = 2.25 GPa, and for the PVC pipe of 63 mm outer diameter and 4.7 mm wall, with Y = 3 GPa and ν = 0.4, the wall term is ε = 10.0, so that the tube wave travels at 453 m/s, and the resonances are spaced by $c/(2L)$ = 227 Hz. As Young's modulus of PVC varies between products and with temperature, these values are estimates, and the speed is measured (Sec. II.D). Eq. (4) assumes a quasi-static wall, which holds well below the ring frequency of the pipe, about 8 kHz.

C. Reciprocity in the fixed tube

With the centers of all three hydrophones at the distance x_0 from their cap, projector P and reciprocal hydrophone T on one cap and the hydrophone H to be calibrated on the other, the paths from P or T to H and the path from P to T have the transfer functions G_f and G_s of Eq. (3) for $(x_s, x_r) = (x_0, L - x_0)$ and (x_0, x_0). The three complex readings of a digital receiver with the gain g in MSB/V, referred to the drive voltage amplitude U, y_{PH}, y_{PT} and y_{TH}, together with the reciprocity of T, give the system sensitivities of H and T in MSB/µPa

$$K_H^2 = J_t \,\frac{g\,Z_T}{U}\,\frac{y_{PH}\,y_{TH}}{y_{PT}}, \qquad K_T = K_H\,\frac{y_{PT}}{y_{PH}}\,\frac{\cos k x_0}{\cos(k(L-x_0))} \tag{5}$$

with the reciprocity parameter of the fixed arrangement

$$J_t = \frac{G_s}{G_f^2} = \mathrm{i}\,\frac{A \sin kL \cos(k(L-x_0))}{\rho c \cos^3 k x_0} \approx \mathrm{i}\,\frac{A \sin 2kL}{2\rho c} \tag{6}$$

where Z_T is the electrical impedance of T and the approximation holds for $x_0 = 0$. All quantities are complex, and J_t has the phase $\pm 90°$. The projector cancels, and neither P nor T needs to be calibrated. For $kL \ll 1$, Eq. (6) approaches $J_t = \mathrm{i}\omega V/(\rho c^2)$, where $V = AL$ is the volume of the pipe, and the pipe acts as a coupler.

D. Calibration frequencies and upper limit of the tube

J_t vanishes at the resonances of the pipe, $f_n = nc/(2L)$, and halfway between them. The calibration is therefore made where $|\sin 2kL|$ reaches its maximum of 1 and does not change to first order with kL, that is, at

$$kL = (2n+1)\,\frac{\pi}{4}, \qquad f = (2n+1)\,\frac{c}{8L} \tag{7}$$

which gives 57, 170, 283 and 396 Hz for the first four. The measured spacing Δf of the resonances gives $kL = \pi f/\Delta f$ independently of c and L, and $c = 2L\Delta f$, which is needed only for the factor $1/c$ of Eq. (6). The uncertain compliance of the wall thereby does not enter the result.

The fixed arrangement has two errors that grow with the frequency. An uncertainty δx_0 of the distance of the hydrophones from the caps changes K_H by the fraction $k\,\delta x_0/2$, and the near field of P at T, estimated as the inertial pressure of P at the distance s from T and of its image in the cap, changes it by $k\ell/2$, with the near-field length

$$\ell = \frac{a^2}{4}\left(\frac{1}{s} + \frac{1}{\sqrt{s^2+4x_0^2}}\right) \tag{8}$$

Combined in quadrature and set equal to the tolerance η, the largest relative error of K_H that is accepted, the two errors give the upper frequency limit

$$f_{\max} \approx \frac{c\,\eta}{\pi\sqrt{\delta x_0^2 + \ell^2}} \tag{9}$$

which is proportional to the tolerance and to the speed of the tube wave, and does not depend on the length of the pipe. It holds only well below $c/(8x_0)$, where the path from P to T has a node at the points $kL \equiv 3\pi/4 \pmod{\pi}$.

E. Reciprocity in the fixed frame

Above the range of the tube, the three-transducer reciprocity is made in the free field (IEC 60565-1, 2020), with the three units fixed at the corners of an equilateral triangle of side d. With the transfer impedance $Z_{ij} = e_j/I_i$, the open-circuit voltage e_j of the receiver j per current I_i into the transmitter i, each of which equals $M_i M_j d_0 e^{-\mathrm{i}kd}/(Jd)$, one obtains

$$M_H^2 = J\,\frac{d}{d_0}\,\frac{Z_{PH}\,Z_{TH}}{Z_{PT}}\,e^{\mathrm{i}kd}, \qquad J = \frac{2d_0}{\rho f} \tag{10}$$

where J is the spherical reciprocity parameter, d_0 = 1 m the reference distance, and $k = 2\pi f/c_w$. The exponential removes the propagation phase, which requires d and c_w: an uncertainty σ_d of each distance and δc of the sound speed cause the phase error

$$\delta\varphi_H = 180^\circ\,\frac{f\,\delta L}{c_w}, \qquad \delta L = \sqrt{3\sigma_d^2 + \left(d\,\frac{\delta c}{c_w}\right)^2} \tag{11}$$

which is linear in frequency and therefore equivalent to a pure delay of $\delta L/(2c_w)$.

The free-field condition holds only until the first unwanted arrival. FM sweeps are deconvolved, and the compressed pulse, which is symmetric about the direct arrival, is gated with a window of length $T_w = d/c_w$ centered on the direct arrival, so that every echo delayed by more than $T_w/2$ is excluded. The echo of the third unit, with the path $2d$, arrives T_w after the direct sound. Echoes delayed by less than $T_w/2$ cannot be gated out. For a tube of the frame, the specular echo relative to the direct sound is estimated with geometric acoustics as

$$\varepsilon = \frac{d}{s_i}\sqrt{\frac{s_i\,\rho_2}{(s_i+s_r)(\rho_2+s_r)}}, \qquad \frac{1}{\rho_2} = \frac{1}{s_i} + \frac{2\sin^2\alpha}{a_F\cos\theta} \tag{12}$$

where s_i and s_r are the distances of the specular point from source and receiver, a_F the radius of the tube, θ the angle of incidence and α the angle between the incident ray and the axis of the tube. Along its axis, the tube acts as a plane mirror, whereas across it the convex surface spreads the reflected wave. Eq. (12) is corrected with the form function of a rigid cylinder at the transverse wavenumber $k\sin\alpha$, with the directivity of both hydrophones at the elevations of the rays, and, for a flooded PVC tube, with the mass-law reflection $|R| = x/\sqrt{1+x^2}$, $x = \omega\rho_w w\cos\theta/(2\rho c_w)$, of a wall of thickness w and density ρ_w, counted for front and back wall. An echo that is the same on all three paths enters Eq. (10) once, so that the error of $|M_H|$ is half that of a single path.

F. Joint estimation

The parameters of Eq. (2) are estimated in two stages. The impedance, measured for each hydrophone, gives C_H, f_k, Q_k and c_k with uncertainties. The acoustic data of tube and frame then determine M_0 and d_c and refine the impedance parameters, by minimizing the sum of the squared residuals

$$r = \frac{\ln(y/M)}{\sigma}, \qquad y_{\mathrm{frame}} = M\,e^{-\mathrm{i}\omega\tau} \tag{13}$$

of the logarithms of the measured complex sensitivities y, weighted with their uncertainties σ, in amplitude and phase, and of the deviations of the impedance parameters from their electrical estimates,

weighted with the uncertainties of these. For the frame, a free delay τ absorbs the linear phase of Eq. (11). The electro-mechanical model thereby fixes the absolute phase at high frequencies, which the geometry of the frame alone determines only up to this delay.

III. MATERIALS AND METHODS

A. Hydrophones, generator and receiver

The transducers are SQ17-00 hydrophones, cylinders of 12.7 mm diameter and 25.4 mm length with a nominal sensitivity of −202.5 dB re 1 V/µPa and a capacitance of 1.6 nF, potted on penetrators. The transmitters are driven by the stereo digital-to-analog converter PCM5122 of Texas Instruments, which, supplied from 3.3 V, delivers a ground-centered full scale of 2.1 V rms into loads of 1 kΩ and more, at sampling rates of up to 384 kHz (Texas Instruments, PCM512x data sheet). The receiver is LC-Mare, with a differential preamplifier of gain 21, an input resistance R_{in} of 4 MΩ in parallel with about 5 pF, and the converter TLV320ADC6140 at 0 dB programmable gain. All readings are taken as complex amplitudes and referred to the drive voltage, which is recorded on a second channel of the receiver through a resistive divider. The ratio of the two channels is then independent of the start of the recording and of the offset between the clocks of generator and receiver. [todo: board used for the PCM5122, firmware]

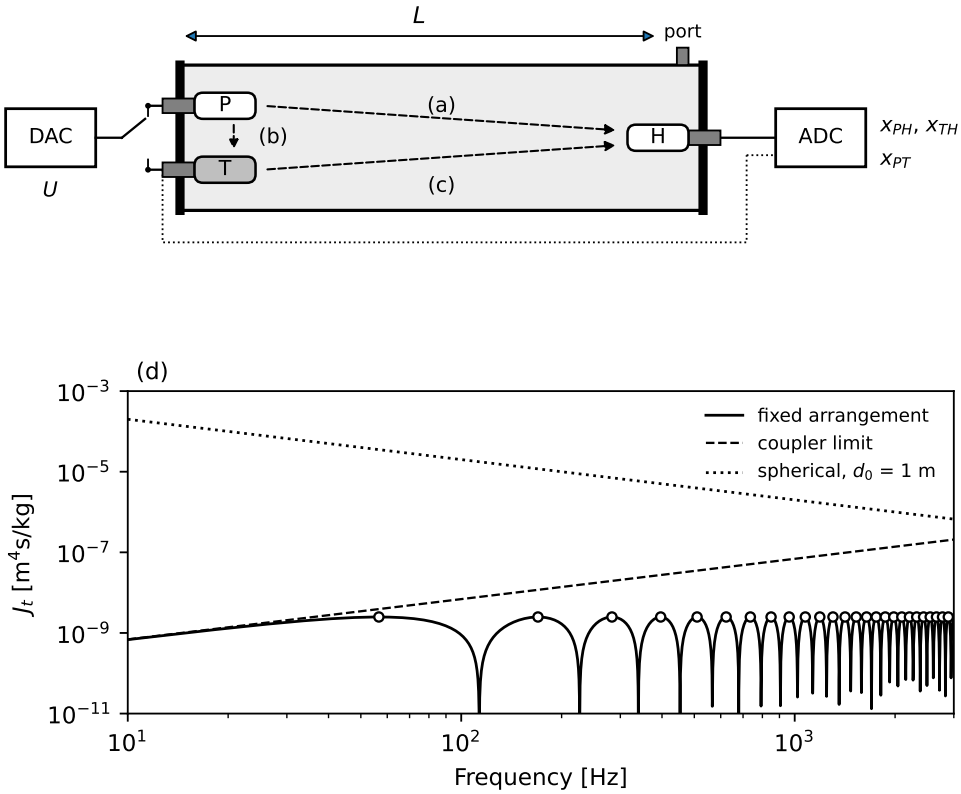


Figure 1: Fixed arrangement in a closed water-filled PVC pipe of length L: projector P and reciprocal hydrophone T (gray) on one end cap, hydrophone H and the filling port on the other, all hydrophones SQ17-00 on penetrators. The converter (DAC) drives P or T with the voltage amplitude U, and the receiver (ADC) reads x_{PH} in measurement (a) and x_{TH} in measurement (c) from H, and x_{PT} in measurement (b) from T (dotted). (d) Magnitude of the reciprocity parameter J_t (solid, Eq. (6) for $x_0 = 0$), of its low-frequency limit $\omega V/(\rho c^2)$ (dashed) and of the spherical reciprocity parameter for d_0 = 1 m (dotted), for a pipe of 1 m length, 63 mm outer diameter and 4.7 mm wall with c = 453 m/s; the circles mark the calibration frequencies of Eq. (7) (c1_pipe.py).

B. Fixed tube

The pipe is a PVC-U pressure pipe of 63 mm outer diameter and 4.7 mm wall [todo: pressure rating, manufacturer], cut to 1 m and closed by two end caps [todo: material, thickness, fixing] (Figure 1). The hydrophones are mounted with their centers at x_0 = 30 mm from the inner faces of the caps, P and T side by side, 25 mm apart, on one cap, and H on the other. The hydrophones are potted in penetrators with M10 thread (Blue Robotics), which are inserted from inside, so that their heads of 18 mm diameter rest on the inner face of the cap, and nuts hold them from outside. Two hydrophones side by side block about 11% of the cross-section, which Eq. (3) neglects. The pipe is filled through a port of 6 mm bore in the cap of H, with this end held highest, so that the air escapes before the port is closed. A trapped bubble of 36 mm³ changes K_H by 0.1 dB. [todo: degassing of the water; temperature]

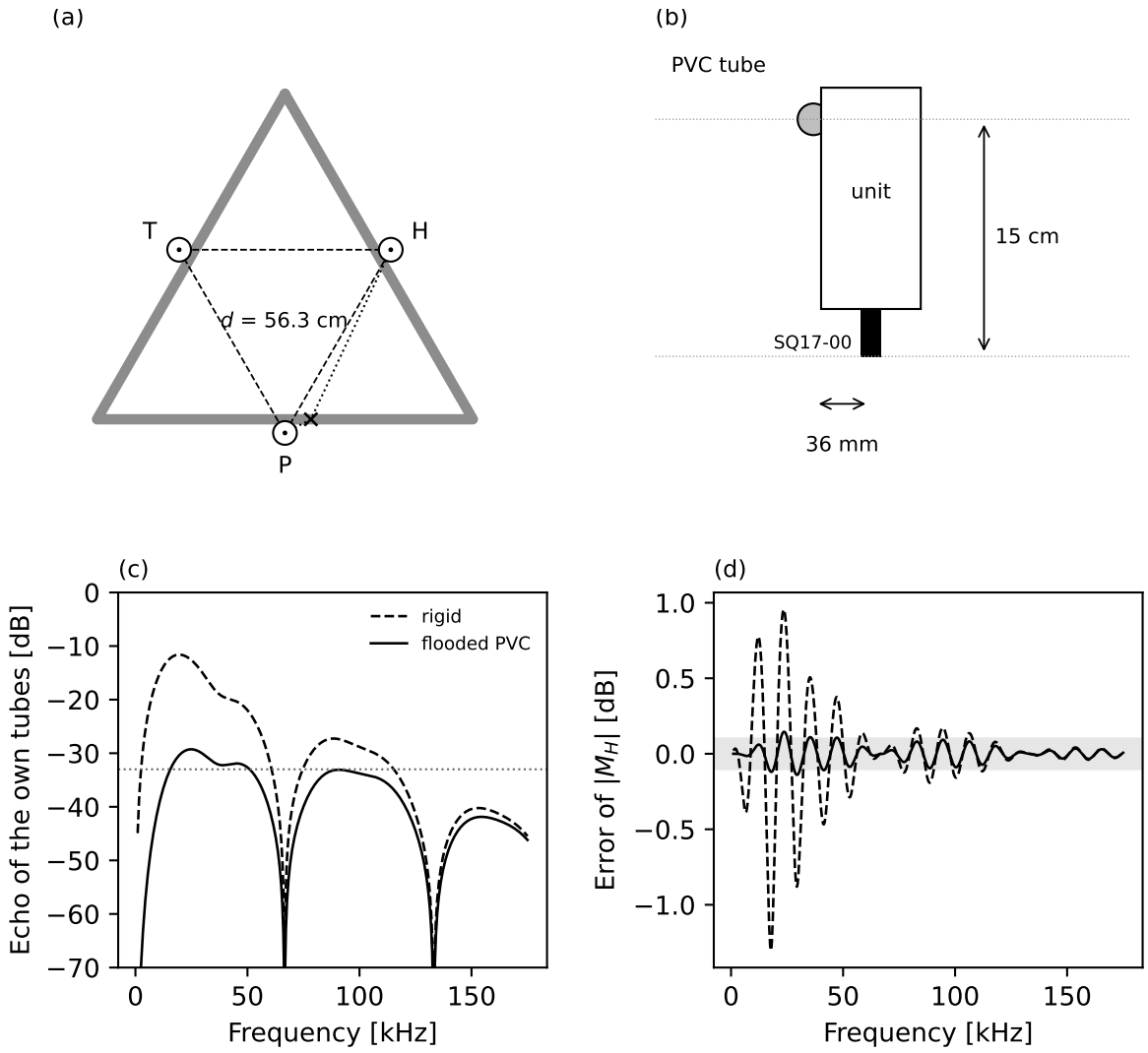


Figure 2: Fixed frame for the free-field reciprocity. (a) Plan: equilateral triangle of flooded PVC tubes of 20 mm diameter and 1 m length (gray), with the units P, T and H hung beside the midpoints of the sides, outside the frame, at the distance d = 56.3 cm from each other (dashed), and the specular point on the own tube of P for the path from P to H (cross, dotted). (b) Elevation, schematic: tube, unit hung upside down, and hydrophone 15 cm below the axis of the tube and 36 mm outward of it. (c) Echo of the own tubes of P and H on the path from P to H relative to the direct sound, for rigid (dashed) and flooded PVC tubes with a wall of 1.5 mm (solid), with the limit of −33 dB for an error of 0.1 dB (dotted), without the shadow of the housing. (d) Resulting error of $|M_H|$ from Eq. (10), with the tolerance ± 0.1 dB (shaded) (c4_frame.py).

C. Fixed frame

The frame is an equilateral triangle of PVC tubes of 20 mm outer diameter and 1 m length, flooded through vent holes so that no air remains trapped, as an air-filled tube would reflect almost totally (Figure 2). The three units, P, T and H, are built into pressure tubes of the same size and hung upside down beside the midpoints of the sides, outside the frame, with the hydrophones on the lower end caps. The hydrophones thereby lie in a horizontal plane 15 cm below the axes of the tubes and 36 mm outward of them, with their axes vertical and parallel, and at the distance d = 56.3 cm from each other. The mounting is mirror-symmetric with respect to the two other units, so that each unit receives from and transmits to both in the same way, as the reciprocity of Eq. (10) requires. The rope is attached with a bridle to a ring well above the frame, and the frame is lowered to a depth of about 10 m.

Two of the units record: H and a second recorder R. In a first session, P transmits and R records T. After recovery, the roles of P and T are exchanged on deck, and in a second session T transmits and R records P. The first session yields Z_{PT} and Z_{PH}, the second Z_{TH} and Z_{TP}. The measurement of Z_{TP} is redundant, as reciprocity requires $Z_{TP} = Z_{PT}$, and the ratio of the two checks that P and T did not change between the sessions. For the phase, the voltage across a series resistor in the drive line is fed to the second channel of both H and R with a cable along the frame, so that each recorder measures its transfer impedances on its own clock. The temperature is logged in each session, as the sound speed enters Eq. (10) and Eq. (11).

D. Workflow

The workflow consists of four steps and an optional comparison (Table 1).

1. **Electrical characterization.** The impedance of each hydrophone is measured in water with a series resistor and two channels of the receiver, from about 10 Hz to 190 kHz. At low frequencies it gives C_H, and near 76 kHz and near 190 kHz the motional branches of the two modes, from which f_k, Q_k and c_k are fitted. The complex response $g_i(f)$ of each receiver channel is measured by replacing the hydrophone by a capacitor equal to its C_H, which captures the high-pass filters of the front end and the differences between channels.
2. **Tube.** The resonances of the filled pipe are measured, and Δf and the calibration frequencies of Eq. (7) are obtained from them. At each calibration frequency, the three acoustic measurements

Table 1: Steps of the workflow. The equipment is the same throughout: the converter PCM5122 as generator, LC-Mare as receiver, a series resistor, a capacitor of about 1.6 nF, the water-filled pipe and the frame.

Step	Measurement	Frequencies	Result
1. Electrical	impedance of each hydrophone in water, response of each receiver channel with a capacitor in place of the hydrophone	10 Hz to 190 kHz	C_H, f_k, Q_k, c_k with uncertainties; complex $g_i(f)$
2. Tube	fixed-arrangement reciprocity in the water-filled pipe	36 to 106 Hz continuously, calibration points up to 736 Hz	complex sensitivity of H and T
3. Frame	fixed-frame reciprocity in the sea, two sessions with P and T exchanged	2.2 to 174 kHz	complex sensitivity of P, T and H, up to a delay
4. Model	joint fit of Eq. (2) to steps 1 to 3 (Eq. (13))	10 Hz to 170 kHz	continuous complex sensitivity of each unit
5. Comparison	optional: further hydrophones side by side with a reference at the cap of H, exchanged	up to 3.0 kHz	M_X/M_R, complex

Table 2: Expected levels in the PVC pipe of 1 m length, 63 mm outer diameter and 4.7 mm wall, with SQ17-00 hydrophones, a drive of 2.1 V rms and LC-Mare as receiver: calibration frequency f, reciprocity parameter J_t for x_0 = 30 mm, impedance of T, sound pressure level p_L at H, reading x_{TH} relative to full scale, and signal-to-noise ratio (SNR) of a Fourier coefficient over 1 s (c1_pipe.py).

kL	f [Hz]	$\lvert J_t \rvert$ [m⁴s/kg]	$\lvert Z_T \rvert$ [kΩ]	p_L [dB re 1 µPa]	x_{TH} [dBFS]	SNR [dB]
$1\pi/4$	57	2.6×10^{-9}	1757	93	−92	28
$3\pi/4$	170	2.3×10^{-9}	586	102	−82	47
$5\pi/4$	283	2.8×10^{-9}	351	106	−77	55
$7\pi/4$	396	2.1×10^{-9}	251	109	−74	61

are made with the same drive, and the impedance of T is measured in its position, with the input admittance of the receiver removed. Eq. (5) is evaluated with Eq. (6) for the actual x_0, and $M_H = K_H/g_H(f)$.

3. **Frame.** The three units are hung in the frame, and the two sessions of Sec. III.C are made with FM sweeps from about 0.5 to 175 kHz. The impulse responses are gated as in Sec. II.E, and Eq. (10) gives the sensitivities of all three units. A repetition with a different drop z checks the echoes of the tubes (Sec. IV.D).
4. **Model.** Eq. (2) is fitted jointly to the impedance and to the sensitivities of steps 2 and 3 (Eq. (13)), for each unit. The residuals of the frame data that the model does not represent, for example the diffraction at the housing, are kept as a correction where they exceed the tolerance.
5. **Comparison.** Further hydrophones may be compared with a reference side by side at the cap of H in the pipe, in two configurations with exchanged positions, which requires no knowledge of the geometry.

IV. RESULTS AND DISCUSSION

A. Expected levels in the tube

At 170 Hz, the drive of 2.1 V rms produces 102 dB re 1 µPa at H, and the reading lies at −82 dBFS, far below clipping (Table 2). The noise is dominated by the thermal noise of R_{in}, which the hydrophone capacitance shorts less and less toward low frequencies, and the signal-to-noise ratio over 1 s is 47 dB. At 57 Hz, it drops to 28 dB, and a ratio of 40 dB requires an averaging over about 14 s. Compared with a rigid tube of the same inner diameter, the elastic wall costs 21 dB of sound pressure at $kL = 3\pi/4$.

B. Error budget and upper frequency limit of the tube

The errors of the fixed arrangement grow with the frequency (Figure 3), as the positions of the hydrophones and the near field of P matter on the scale of the wavelength. At 170 Hz, the residual errors add up to about 0.14 dB (Table 3). At 57 Hz, the noise dominates and is reduced by a longer averaging, whereas above 170 Hz the near field of P at T dominates. For the distance of 25 mm between P and T, Eq. (8) gives ℓ = 9.9 mm. A tolerance of 0.5 dB, from which an error of 1% in c and of 0.1 dB in $\lvert Z_T \rvert$ are deducted in quadrature, corresponds to η = 0.057, and with δx_0 = 5 mm, Eq. (9) gives 739 Hz. The highest calibration point with errors below 0.5 dB is 736 Hz, where kx_0 = 0.31 rad, so that the first-order estimate holds well. The points $kL \equiv 3\pi/4$ fail earlier: at 623 Hz, the near field alone causes 0.51 dB, as the path from P to T weakens toward its node. Moving P and T to the largest distance that the heads of the penetrators leave, 36 mm, would shorten ℓ to 7.6 mm and raise the estimate to 903 Hz.

Larger tolerances raise the limit, but much less than in proportion (Table 4). Beyond about 1 dB, the limit is set by the node of the path from P to T at $c/(8x_0)$ = 1.9 kHz and by the node of H at $c/(4x_0)$ = 3.8 kHz. A large tolerance is nevertheless of little use, because the frame takes over at higher frequencies and the model joins the two.

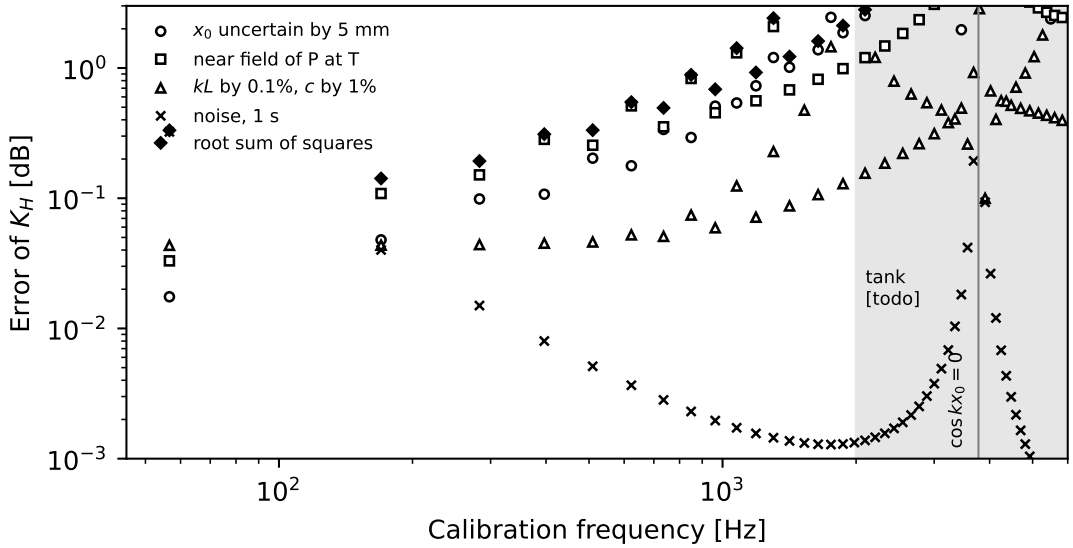


Figure 3: Errors of the system sensitivity K_H as a function of the calibration frequency of Eq. (7) for the pipe of Table 2: an error of 5 mm in x_0 (circles), the near field of P at T (squares), an uncertainty of 0.1% in kL and of 1% in c (triangles), the noise of a Fourier coefficient over 1 s (crosses), and the root sum of squares of these and of an error of 0.1 dB in $|Z_T|$ (diamonds). The vertical line marks $\cos kx_0 = 0$ (c1_pipe.py).

It is evident from Figure 4(a) that the errors of x_0 and of the near field alternate in sign from one calibration point to the next, as the transfer function G_s of the path from P to T alternates in sign with $\tan kL$. The mean half-difference of a point to its two neighbors is therefore of the order of its systematic error (Figure 4(b)), and the weighted average $(K_{n-1} + 2K_n + K_{n+1})/4$ of three neighbors, taken in decibels, cancels the part that alternates in sign: it reduces the error from 0.16 to 0.01 dB at 170 Hz and from 0.39 to 0.06 dB at 396 Hz. In the joint fit of step 4, all calibration points enter with their own uncertainty, which achieves a similar averaging. The phase of the tube is affected only by the losses of the wall. For a loss factor of 0.02 of PVC [todo: verify], the phase errors remain below 0.5° and the amplitude errors below 0.01 dB at the first three calibration points.

Table 3: Error budget of K_H in dB at the first four calibration frequencies. The errors that are corrected are listed for their size but excluded from the root sum of squares (c1_pipe.py).

Source of error	57 Hz	170 Hz	283 Hz	396 Hz
x_0 ignored (corrected by Eq. (6))	−0.10	0.30	−0.55	0.67
x_0 uncertain by 5 mm	0.02	0.05	0.10	0.11
kL uncertain by 0.1%, c by 1%	0.04	0.04	0.04	0.05
near field of P at T (estimate)	0.03	0.11	0.15	0.28
loading of T by the receiver (corrected)	0.39	0.06	0.03	0.02
$\lvert Z_T \rvert$ uncertain by 0.1 dB	0.05	0.05	0.05	0.05
noise, 1 s	0.32	0.04	0.01	0.01
root sum of squares of the residual errors	0.33	0.14	0.19	0.31

Table 4: Upper frequency limit of the pipe for several tolerances of K_H: estimate of Eq. (9) and highest calibration point below $c/(4x_0)$ whose residual errors (Table 3) remain within the tolerance (c1_pipe.py).

Tolerance	0.5 dB	1 dB	3 dB	5 dB
$f_{\max}$, Eq. (9)	739 Hz	1488 Hz	4473 Hz	7457 Hz
highest calibration point	736 Hz	1189 Hz	2095 Hz	2322 Hz

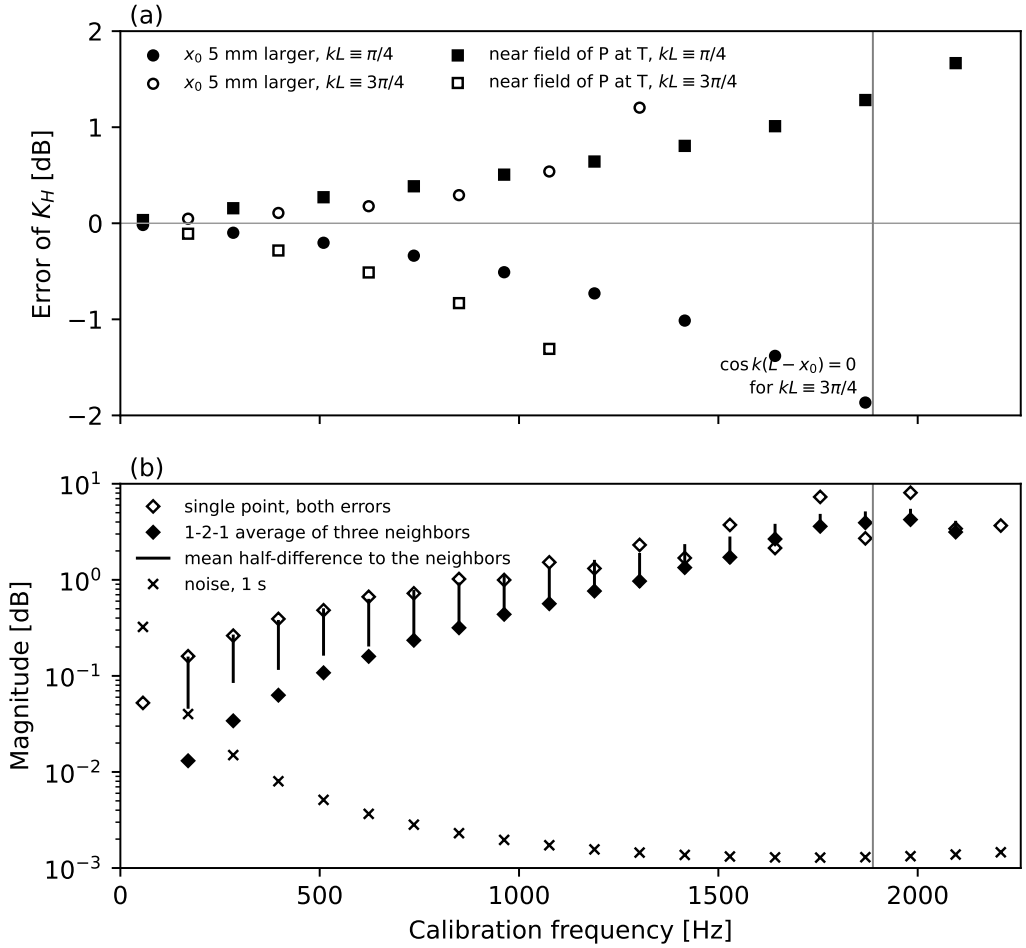


Figure 4: Signed errors of K_H at the calibration points of Eq. (7). (a) Error for hydrophones 5 mm farther from the caps than assumed (circles) and estimated error of the near field of P at T (squares), at the points $kL \equiv \pi/4$ (mod π, filled) and $kL \equiv 3\pi/4$ (mod π, open); the vertical line marks $f = c/(8x_0)$. (b) Magnitude of the sum of both errors at a single point (open diamonds) and of the 1-2-1 average of three neighboring points (filled diamonds), each for the larger of the results for x_0 errors of +5 and −5 mm, the range of the mean half-difference of a point to its two neighbors (vertical bars), and the noise over 1 s (crosses) (c1_pipe.py).

Table 5: Free-field levels in the frame for SQ17-00 hydrophones as projector and reciprocal hydrophone, driven with 2.1 V rms, at d = 56.3 cm, with LC-Mare as receiver: sound pressure level p_L at H, equivalent self-noise N_L of the receiver, signal-to-noise ratio of a tone burst gated to T_w = 0.375 ms, spherical reciprocity parameter J, and ka of the housing of radius 31.5 mm (c4_frame.py).

f [kHz]	p_L [dB re 1 µPa]	N_L [dB re 1 µPa²/Hz]	SNR [dB]	J [m⁴s/kg]	ka housing
2	80	37	5	1.0×10^{-6}	0.3
5	95	37	22	4.0×10^{-7}	0.7
10	107	36	34	2.0×10^{-7}	1.3
20	120	36	46	1.0×10^{-7}	2.6
50	136	36	62	4.0×10^{-8}	6.6
100	146	38	71	2.0×10^{-8}	13.2
130	152	37	78	1.5×10^{-8}	17.2

C. Levels, gate and sweeps in the frame

Below its first resonance, the SQ17-00 radiates a level that rises by about 12 dB per octave (Table 5), from 95 dB re 1 µPa at 5 kHz to 136 dB re 1 µPa at 50 kHz at 56.3 cm. At the lower end of the frame, the signal-to-noise ratio of a single gated burst is small, 5 dB at 2 kHz against the self-noise of the receiver alone, and less against the ambient noise of the sea [todo: ambient noise at the site]. A sweep of duration T_s over the bandwidth B carries the energy density $A^2 T_s/(2B)$, which exceeds that of a burst gated to T_w by the factor $T_s/(BT_w^2)$, about 16 dB for a sweep of 1 s over 170 kHz, and N sweeps add $10 \log_{10} N$ dB.

The gate of length T_w = 0.375 ms extends 188 µs to either side of the direct arrival. At 10 m depth, the surface echo arrives 13.0 ms after the direct sound and is irrelevant. Within the frame, the echo of the third unit arrives after 375 µs, that of the opposite tube after 361 µs and that of the nearest corner after 322 µs, all outside the gate. The gate smooths the response over $1/T_w$ = 2.7 kHz. A single sweep from 0.5 to 175 kHz with raised-cosine edges of width $1/T_w$, including the echo of the third unit at its geometric upper estimate of −19 dB and that of the opposite tube, is usable within 0.1 dB and 1° from 2.2 kHz upward. When the band is split into sub-bands, each loses part of its band at both edges, as the gate truncates the compressed pulse. The minimal bandwidth is about $4/T_w$ = 10.7 kHz, of which 4.8 kHz are usable, and doubling it to 21.3 kHz raises the usable band to 16.5 kHz, from 0.45 to 0.77 of the band. The amplitude is limited by the passband of the decimation filter of the converter in the two-channel mode at 384 kHz, about 174 kHz.

D. Echoes of the frame

With the units hung 15 cm below the frame, the tubes lie close to the direct paths. For the path from P to H, the own tube of P and that of H each have a specular point 6.9 cm from the midpoint of the side, with an excess path of 12.7 cm and a delay of 85 µs, far within the half gate. These echoes cannot be gated out and cause a ripple with a period of 12 kHz. For rigid tubes, Eq. (12) gives a combined echo of up to −12 dB, which would bias $|M_H|$ by up to 1.30 dB (Figure 2(c, d)). The flooded PVC tubes reduce it to −29 dB near 25 kHz, because the impedance of PVC is only about twice that of water and the wall is thin, and because the wave meets the tube at a steep angle. The bias of $|M_H|$ then stays below 0.15 dB and 1.0°, close to the tolerance. This estimate is an upper one. The ray from the hydrophone to its own tube rises at 63° and leaves the radius of the housing only 6.1 cm above the hydrophone, so that the housing shadows it unless the hydrophone protrudes further below the end cap [todo: protrusion of the SQ17-00 below the end cap]. The check $Z_{TP} = Z_{PT}$ does not reveal these echoes, as they are the same in both sessions, whereas a repetition with a different drop z changes their delay and their ripple but not the sensitivity of the units.

For the phase, Eq. (11) gives δL = 0.9 mm for σ_d = 0.5 mm and δc = 0.5 m/s, and 1.9 mm for 1.0 mm and 2.0 m/s, so that the absolute phase from the geometry alone is within 5° only up to 47 or 22 kHz. The error is, however, a pure delay of 0.3 to 0.6 µs, less than the sampling interval of 2.6 µs, and the delay τ of Eq. (13) absorbs it.

E. Joint estimation

The benefit of the combination was simulated (c5_workflow.py). In each of 200 realizations, the true parameters of a hydrophone are drawn around those of the SQ17-00 model, with M_0 uncertain by 3 dB before calibration and d_c by 30%. The electrical fit gives the impedance parameters with errors of 1% in f_1, 2% in f_2, 10% in the quality factors and 5% in the coupling factors. The tube provides the calibration points up to 736 Hz with the uncertainties of Figure 3 and 1° in phase. The frame provides the sensitivity from 2.2 kHz upward, biased by the echo of the own tubes of Figure 2(d), with a random error of 0.05 dB and 0.3°, and delayed by an unknown τ with a standard deviation of 0.3 µs. The model is then fitted to impedance and tube alone, or to all three (Eq. (13)).

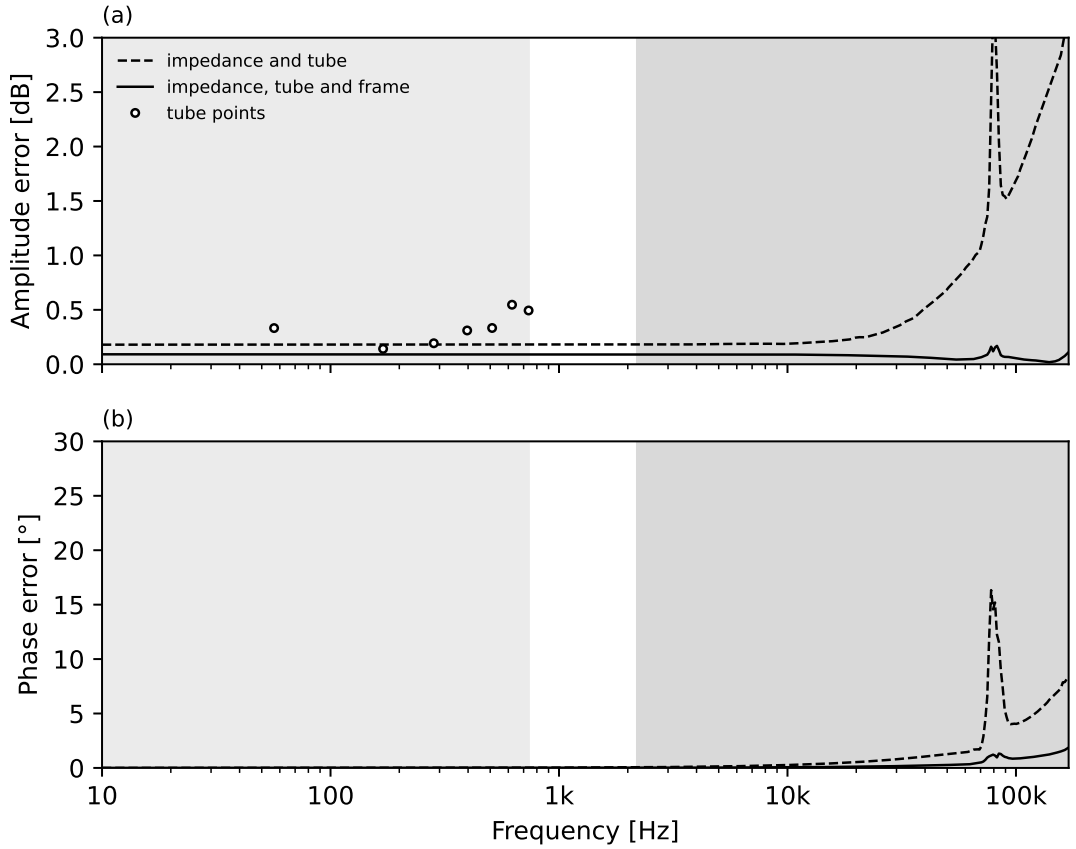


Figure 5: Errors of the fitted electro-mechanical model, 95th percentile of 200 Monte Carlo realizations, in (a) amplitude and (b) phase: model fitted to impedance and tube (dashed) and to impedance, tube and frame (solid), and the uncertainty of the tube points (circles). The light band marks the calibration points of the tube up to 736 Hz, the dark band the usable range of the frame from 2.2 kHz (c5_workflow.py).

It is evident from Figure 5 and Table 6 that the impedance and the tube determine the response well below a few tens of kilohertz, where the hydrophone is flat and only M_0 matters, but leave errors of up to 3.26 dB and 16.3° near the first resonance and up to 3.13 dB above 100 kHz, where d_c and the quality factors dominate. With the frame, the errors drop to 0.17 dB and 1.3° near the resonance and to 0.09 dB and 1.8° at the top of the band. The frame data also improve the low frequencies, from 0.18 to 0.09 dB, as M_0 is common to all frequencies. The ripple of the tube echoes, with a period of 12 kHz, cannot be represented by Eq. (2) and is largely averaged out by the fit. The gap between 736 Hz and 2.2 kHz, where neither measurement applies, is bridged by the model with the accuracy of M_0.

These numbers assume that the response of the unit follows Eq. (2). The housing of the recorder, with ka above 1 from about 8 kHz, adds a diffraction that the circuit does not describe, and its residual in

Table 6: Largest 95th percentile of the errors of the fitted model by frequency band (c5_workflow.py). The phase of the model at low frequencies is that of the circuit and is not affected by the fit.

	impedance and tube		impedance, tube and frame	
Band	amplitude [dB]	phase [°]	amplitude [dB]	phase [°]
10 Hz to 736 Hz	0.18	0.0	0.09	0.0
736 Hz to 2.2 kHz	0.18	0.1	0.09	0.0
2.2 to 50 kHz	0.68	1.3	0.09	0.3
50 to 100 kHz	3.26	16.3	0.17	1.3
100 to 170 kHz	3.13	8.1	0.09	1.8

Table 7: Comparison of the fixed tube and the fixed frame.

	Fixed tube	Fixed frame
Field	plane waves in a coupler	free field, gated
Frequencies	36 to 736 Hz	2.2 to 174 kHz
Signals	tones at the calibration points	FM sweeps, deconvolved and gated
Mounting	hydrophone on its penetrator	complete unit, as deployed
Main limitations	x_0, near field, tube-wave speed, air	echoes of the tubes, ambient noise, distances and sound speed for the phase
Phase	from J_t, within about 1°	up to a delay, fixed by the model
Cost	[todo:]	[todo:]

the frame data is kept as a correction (step 4). As this residual and the ripple of the tube echoes have similar periods, they are separated only by the repetition with a different drop, and by the low level of the echoes of the flooded tubes. Below 36 Hz, the noise of the receiver dominates the tube: at 10 Hz, a coefficient over 1 s has an error of 0.85 dB and 5.6°, which an averaging over 10 s lowers to 0.28 dB and 1.8°.

F. Comparison of further hydrophones

With the exchange of positions and a reproducibility of the mounting of 0.5 mm, the comparison of a further hydrophone with a reference at the cap of H adds less than 0.1 dB and 1° up to 3.0 kHz, close to the node of H at $c/(4x_0)$ = 3.8 kHz. Without the exchange, a difference of 2 mm between the positions would cause 0.27 dB at 1.5 kHz. The comparison is therefore suited to calibrating a batch of hydrophones at low frequencies, whereas the frame calibrates three complete units per deployment.

Table 7 compares the two measurements. Both are reciprocity methods with fixed arrangements, and their traceability rests on electrical quantities only, the drive voltage, the receiver gain and the impedance of the reciprocal transducer. The tube covers the low frequencies that a free field cannot reach, with the hydrophone on its penetrator, and the frame covers the high frequencies with the complete unit, including its housing. The model joins them and is in turn tested by them.

G. Comparison with an in-tank calibration

[todo: Results: K_H of tube and frame; K_H of a tank calibration of the same recorder by comparison with a reference hydrophone; M_H = K_H/g(f) of all on a common axis; differences and combined uncertainties.]

A systematic difference that is the same at all frequencies would point to the electrical calibration, that is, to g, U or $|Z_T|$, whereas a difference that grows with the frequency would point to the near field or to x_0 in the tube, or to the echoes of the frame.

V. CONCLUSIONS

A calibration workflow for the low-cost recorder LC-Mare with SQ17-00 hydrophones was presented that combines an electro-mechanical model of the hydrophone with two reciprocity measurements in fixed arrangements. The impedance gives the shape of the response. A water-filled PVC pipe with three hydrophones on penetrators gives its level from 36 to 736 Hz, at calibration frequencies obtained from the measured resonances, with residual errors of 0.14 dB at 170 Hz and a limit set by the near field of P at T and the positions of the hydrophones. A frame of flooded 20 mm PVC tubes, with the three units hung 15 cm below it at 56.3 cm from each other and lowered to 10 m, gives the free-field sensitivity of the complete units from 2.2 to 174 kHz with gated FM sweeps, provided the sub-bands are at least 10.7 kHz wide. Its weak point is the echo of the own tubes, which falls within the gate, but which the flooded PVC tubes keep below 0.15 dB. The joint fit of the model bridges the gap between the two

measurements, fixes the absolute phase that the geometry of the frame leaves open, and reduces the expected errors to about 0.17 dB and 1.8° over the band, provided the response follows the model. The tube is therefore preferred for the low frequencies and the frame for the high frequencies, whereas the model alone, fitted to impedance and tube, should not be trusted near and above the first resonance. [todo: Conclusion of the comparison with the in-tank calibration.]

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